



习题 2.32

$$\begin{cases} x^2 y'' + xy' + (x^2 - m^2)y = 0, \quad m > 0 \\ y = x^c \sum_{n=0}^{\infty} a_n x^n \end{cases}$$

$$\Rightarrow x^2 \sum_{n=0}^{\infty} (c+n)(c+n-1) a_n x^{c+n-2} + x \sum_{n=0}^{\infty} (c+n) a_n x^{c+n-1} + x^2 \sum_{n=0}^{\infty} a_n x^{n+c} - m^2 \sum_{n=0}^{\infty} a_n x^{n+c} = 0$$

$$\Leftrightarrow \sum_{n=0}^{\infty} (c+n)(c+n-1) a_n x^{n+c} + \sum_{n=0}^{\infty} (c+n) a_n x^{n+c} + \sum_{n=0}^{\infty} a_n x^{n+2+c} - \sum_{n=0}^{\infty} m^2 a_n x^{n+c} = 0$$

$$\Leftrightarrow \sum_{n=0}^{\infty} [(c+n)^2 a_n - m^2 a_n] x^{n+c} + \sum_{n=0}^{\infty} a_n x^{n+2+c} = 0$$

$$(c^2 - m^2) a_0 = 0$$

$$[c(c+1)^2 - m^2] a_1 = 0$$

$$[(c+n)^2 - m^2] a_n + a_{n-2} = 0, \quad n \geq 2.$$

(i) 当 $c = m-1$ 时, ~~$a_0 = 0$~~ 有 $a_0 = 0 \Rightarrow a_{2k} = 0, \quad k \in \mathbb{N}$

~~a_{2k+3}~~

$$a_{2k+3} = - \frac{a_{2k+1}}{[(m-1+2k+2)^2 - m^2]} = - \frac{a_{2k+1}}{(2k+2)(2m+2k+2)} = - \frac{a_{2k+1}}{4k(m+k)}, \quad k \in \mathbb{N}^0$$

$$= \dots = (-1)^{k+1} \frac{a_1}{4^{k+1} (k+1)! (m+k+1)(m+k) \dots (m+1)}$$

$$\text{记 } a_1 = \frac{b_1}{I(m+1)}$$

$$a_{2k+3} = \frac{(-1)^{k+1} b_1}{4^{k+1} (k+1)! I(m+k+2)}$$

$$J_m(x) = y(x) = x^{m-1} \sum_{k=0}^{\infty} a_{2k+1} x^{2k+1} = x^{m-1} \left[a_1 x + \sum_{k=0}^{\infty} a_{2k+3} x^{2k+3} \right]$$

$$= x^{m-1} \left[a_1 x + \sum_{k=0}^{\infty} \frac{(-1)^{k+1} b_1}{4^{k+1} (k+1)! I(m+k+2)} x^{2k+3} \right]$$

$$= x^m \sum_{k=0}^{\infty} \frac{(-1)^k b_1}{4^k k! I(m+k+1)} x^{2k} = \sum_{k=0}^{\infty} \frac{(-1)^k b_1 \cdot 2^m}{k! I(m+k+1)} \left(\frac{x}{2} \right)^{m+2k}$$

$$\text{取 } b_1 = 2^{-m}, \text{ 得 } y(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! I(m+k+1)} \left(\frac{x}{2} \right)^{m+2k}$$

$$(4) G(s)H(s) = -\frac{s+1}{s(s-1)} \quad \text{and} \quad \mathcal{L}G(j\omega)H(j\omega) = \frac{2\operatorname{arctan}\omega}{j\omega}$$

(2) 當 $c = -m$ 時, 有 $a_{2k+1} = 0, k \in \mathbb{N}$

$$a_{2k+2} = -\frac{a_{2k}}{[-m+2k+1]^2 - m^2} = -\frac{a_{2k}}{(2k+2)(-2m+2k+2)} = \frac{a_{2k}}{(2k+2)(m-2k-2)} = \frac{a_{2k}}{4(k+1)(m-k-1)}$$

$$= -\frac{a_{2k}}{4(k+1)(-m+k+1)}$$

$$\Rightarrow \text{若 } J_{-m}(x) = y(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(-m+k+1)} \left(\frac{x}{2}\right)^{-m+2k}$$

(5) 習題 2.33

$$\begin{aligned} \frac{d}{dx}(x^m J_m(x)) &= \frac{d}{dx} \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(m+k+1)} \frac{x^{2m+2k}}{2^{m+2k}} \right) = \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(m+k+1)} \cdot \frac{2(m+k) x^{2m+2k-1}}{2^{m+2k}} = \\ &= \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(m+k)} \cdot \frac{x^{2m+2k}}{2^{m+2k}} \right) \cdot x^m \\ &= x^m J_{m-1}(x) \quad \dots \dots \dots (1) \end{aligned}$$

$$\begin{aligned} \frac{d}{dx}(x^{-m} J_m(x)) &= \frac{d}{dx} \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(m+k+1)} \cdot \frac{x^{2k}}{2^{m+2k}} \right) = \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(k-1)! \Gamma(m+k+1)} \cdot \frac{x^{2k-1}}{2^{m+2k-1}} = \\ &= \sum_{k=0}^{\infty} \frac{(-1)^{k+1}}{k! \Gamma(m+1+k+1)} \cdot \frac{x^{2k+1+m-m}}{2^{m+1+2k}} \\ &= -x^{-m} J_{m+1}(x) \quad \dots \dots \dots (2) \end{aligned}$$

由(1)得

$$x^m J_m'(x) + m x^{m-1} J_m(x) = x^m J_{m-1}(x) \Leftrightarrow J_m'(x) + m x^{-1} J_m(x) = J_{m-1}(x)$$

由(2)得

$$x^{-m} J_m'(x) - m x^{-m-1} J_m(x) = -x^{-m} J_{m+1}(x) \Leftrightarrow J_m'(x) - m x^{-1} J_m(x) = -J_{m+1}(x)$$

$$\Rightarrow \begin{cases} J_{m-1}(x) + J_{m+1}(x) = 2m x^{-1} J_m(x) \\ J_{m-1}(x) - J_{m+1}(x) = 2J_m'(x) \end{cases}$$

習題 2.34

當 $J_m(x) = \frac{y}{x}$ 時, 則

$$x^2 \frac{d^2}{dx^2} \left(\frac{y}{x} \right) + x \frac{d}{dx} \left(\frac{y}{x} \right) + (x^2 - m^2) \left(\frac{y}{x} \right) = 0$$

$$\Leftrightarrow x^2 \frac{d}{dx} \left(\frac{y'x - y}{x^2} \right) + x \cdot \frac{y'x - y}{x^2} + (x^2 - m^2) \left(\frac{y}{x} \right) = 0$$

$$\Leftrightarrow x^2 \left(\frac{y''x - y'}{x^2} - \frac{y'x^2 - 2xy}{x^4} \right) + (y' - \frac{y}{x}) + xy - m^2 \frac{y}{x} = 0$$

$$\Leftrightarrow xy'' - y' + (1 - m^2 + x^2) \frac{y}{x} = 0 \Leftrightarrow x^2 y'' - x y' + (1 + x^2 - m^2) y = 0$$



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習題 2.35

(1) 利用遞推式 $J_{m-1}(x) + J_{m+1}(x) = 2mx^{-1}J_m(x) \Leftrightarrow J_{m+1}(x) = 2mx^{-1}J_m(x) - J_{m-1}(x)$

得 $J_2(x) = 2x^{-1}J_1(x) - J_0(x)$

$$J_3(x) = 2 \times 2x^{-1}J_2(x) - J_1(x)$$

$$= 4x^{-1}(2x^{-1}J_1(x) - J_0(x)) - J_1(x) = (8x^{-2} - 1)J_1(x) - 4x^{-1}J_0(x)$$

$$J_4(x) = 2 \times 3x^{-1}J_3(x) - J_2(x)$$

$$= 6x^{-1}[(8x^{-2} - 1)J_1(x) - 4x^{-1}J_0(x)] - 2x^{-1}J_1(x) + J_0(x)$$

$$= (48x^{-3} - 8x^{-1})J_1(x) - (24x^{-2} - 1)J_0(x)$$

(2) ~~$\int x^4 J_1(x) dx = \int x^3 (x^1 J_1(x)) dx = \int x^2 d(x)$~~

$$\int x^4 J_1(x) dx = \int x^4 (x^{-0} J_1(x)) dx = -\int x^4 d(J_0(x))$$

$$= -[x^4 J_0(x) - \int J_0(x) \cdot 4x^3 dx]$$

$$= -x^4 J_0(x) + 4 \int x^3 (x J_0(x)) dx$$

$$= -x^4 J_0(x) + 4 \int x^2 d(x J_1(x)) = -x^4 J_0(x) + 4 \left(x^3 J_1(x) - \int x J_1(x) \cdot 2x dx \right)$$

$$= -x^4 J_0(x) + 4x^3 J_1(x) - 8 \int x^2 (x^{-0} J_1(x)) dx = -x^4 J_0(x) + 4x^3 J_1(x) - 8 \int x^2 d(J_0(x))$$

$$= -x^4 J_0(x) + 4x^3 J_1(x) - 8 \left(x^2 J_0(x) - \int 2x J_0(x) dx \right)$$

$$= -x^4 J_0(x) + 4x^3 J_1(x) - 8x^2 J_0(x) + 16 \int x J_0(x) dx$$

(3) 利用遞推式 $J_{m-1}(x) - J_{m+1}(x) = 2J'_m(x)$

$$J'_3(x) = \frac{1}{2} (J_2(x) - J_4(x)) = \frac{1}{2} [2x^{-1}J_1(x) - J_0(x) - (48x^{-3} - 8x^{-1})J_1(x) + (24x^{-2} - 1)J_0(x)]$$

$$= (5x^{-1} - 24x^{-3})J_1(x) + (12x^{-2} - 1)J_0(x)$$

(1) ~~$J_{m-1}(x) = \frac{1}{x} J_m'(x) - J_m(x)$~~

$$x^{-1} J_2(x) = -\frac{d}{dx} (x^{-1} J_1(x)) = -\frac{1}{x} J_1'(x) + \frac{1}{x^2} J_1(x)$$

$$x^0 J_1(x) = -\frac{d}{dx} (J_0(x)) = -J_0'(x) \Rightarrow J_1'(x) = -J_0''(x)$$

$$\Rightarrow x^{-1} J_2(x) = \frac{1}{x} J_0''(x) - \frac{1}{x^2} J_0'(x) \Leftrightarrow J_2(x) = J_0''(x) - \frac{1}{x} J_0'(x)$$

(2) ~~$$\begin{aligned} J_3(x) &= 2 \times 2 x^{-1} J_2(x) - J_1(x) \\ &= 4 x^{-1} J_2(x) - J_1(x) \\ &= 4 x^{-1} (J_0''(x) - x^{-1} J_0'(x)) - J_0'(x) \\ &= 4 x^{-1} J_0''(x) - 4 x^{-2} J_0'(x) - J_0'(x) \end{aligned}$$~~

~~$$\begin{aligned} x^{-2} J_3(x) &= -\frac{d}{dx} (x^{-2} J_2(x)) = -x^{-2} J_2'(x) + 2 x^{-3} J_2(x) \\ &= -x^{-2} (J_0'''(x) - x^{-1} J_0''(x)) + 2 x^{-3} (J_0''(x) - x^{-1} J_0'(x)) \\ &= -x^{-2} J_0'''(x) + 3 x^{-4} J_0''(x) - 2 x^{-4} J_0'(x) \end{aligned}$$~~

~~$J_2(x)$~~

$$x^{-2} J_3(x) = -\frac{d}{dx} (x^{-2} J_2(x)) = -[x^{-2} J_2'(x) - 2 x^{-3} J_2(x)]$$

$$J_2(x) = -J_2'(x) + 2 x^{-1} J_2(x)$$

$$= (J_1'(x) - x^{-1} J_1(x))' - 2 x^{-1} (J_1'(x) - x^{-1} J_1(x))$$

$$= J_1''(x) - x^{-1} J_1'(x) + x^{-2} J_1(x) - 2 x^{-1} J_1'(x) + 2 x^{-2} J_1(x)$$

$$= J_1''(x) - 3 x^{-1} J_1'(x) + 3 x^{-2} J_1(x)$$

~~$x^{-2} J_1(x) =$~~

$x^{-1} J_1(x)$

$$\begin{aligned}
 (2) J_3(x) &= 4x^{-1} J_2(x) - J_1(x) \\
 &= 4x^{-1} (J_0''(x) - x^{-1} J_0'(x)) + J_0'(x) \\
 &= 4x^{-1} J_0''(x) - 4x^{-2} J_0'(x) + J_0'(x)
 \end{aligned}$$

$$\begin{aligned}
 J_2(x) &= 2x^{-1} J_1(x) - J_0'(x) \\
 &= 2x^{-1} (J_0''(x) - x^{-1} J_0'(x)) - J_0''(x) + x^{-1} J_0'(x) - x^{-2} J_0'(x) \\
 &= -J_0''(x) + 3x^{-1} J_0'(x) - 3x^{-2} J_0'(x)
 \end{aligned}$$

$$\begin{aligned}
 J_1(x) &= 4J_3(x) - 3J_2(x) = 4(-J_0''(x) + 3x^{-1} J_0'(x) - 3x^{-2} J_0'(x)) \\
 &\quad - 3(4x^{-1} J_0''(x) - 4x^{-2} J_0'(x) + J_0'(x)) \\
 &= -4J_0''(x) - 3J_0'(x)
 \end{aligned}$$

$$\Leftrightarrow J_1(x) + 3J_0'(x) + 4J_0''(x) = 0$$

習題 2.37

$$\begin{aligned}
 \int x^n J_0(x) dx &= \int x^{n-1} (x J_0(x)) dx = x^n J_1(x) - (n-1) \int x^{n-1} J_1(x) dx = x^n J_1(x) + \frac{1}{2}(n-1) \int x^{n-1} (-x^0 J_1(x)) dx \\
 &= x^n J_1(x) + (n-1) \left(x^{n-1} J_0(x) - \int J_0(x) \cdot (n-1) x^{n-2} dx \right) \\
 &= x^n J_1(x) + (n-1) x^{n-1} J_0(x) - (n-1)^2 \int x^{n-2} J_0(x) dx
 \end{aligned}$$

習題 2.38

原方程變為 $\frac{1}{y} \frac{d}{dx} \left(k(x) \frac{dy}{dx} \right) - q(x) + \lambda p(x) = 0$

積分後 $\int_0^1 \frac{1}{y} \frac{d}{dx} \left(k(x) \frac{dy}{dx} \right) dx - \int_0^1 q(x) dx + \lambda \int_0^1 p(x) dx = 0$

$$\underbrace{\left. \frac{1}{y} k(x) \frac{dy}{dx} \right|_0^1}_{(*)} - \int_0^1 k(x) \left(\frac{dy}{dx} \right)^2 dx - \int_0^1 q(x) dx + \lambda \int_0^1 p(x) dx = 0$$

結合已知條件知 $(*) = 0$.

而 $k(x), q(x), p(x) \geq 0$, 故 $\int_0^1 k(x) \left(\frac{dy}{dx} \right)^2 dx, \int_0^1 q(x) dx, \int_0^1 p(x) dx \geq 0$.

$$\lambda = \frac{\int_0^1 k(x) \left(\frac{dy}{dx} \right)^2 dx + \int_0^1 q(x) dx}{\int_0^1 p(x) dx} \geq 0$$